

Министерство образования Республики Беларусь

Учреждение образования
«Белорусский государственный университет информатики и
радиоэлектроники»

Кафедра электронных вычислительных машин

Дисциплина: Базы данных

Лабораторная работа №3

РЕАЛИЗАЦИЯ SQL-ЗАПРОСОВ ДЛЯ СОЗДАНИЯ СХЕМЫ БАЗЫ
ДАННЫХ

Вариант № 26: СТО

Выполнил
студент группы 150503:
Шарай П.Ю.

Проверила:
Игнатович А.О.

Минск 2024

1 ЦЕЛЬ

Цель лабораторной работы – реализация схемы базы данных по ранее построенной реляционной схеме данных (см. лабораторную работу №2). Требуется сформировать SQL-запросы для создания таблиц базы данных и выполнить их в СУБД. Требуется заполнить таблицы данными с помощью оператора INSERT.

2 ПОРЯДОК ВЫПОЛНЕНИЯ РАБОТЫ

1. Создать в СУБД новую схему данных для хранения пользовательских объектов;

2. В этой новой схеме данных с помощью скрипта с запросами на языке DDL SQL реализовать таблицы, соответствующие реляционным отношениям схемы данных полученной в лабораторной работе №2, с помощью одного (желательно) оператора CREATE TABLE для каждой таблицы в следующем порядке:

- (a) реализовать простую структуру таблиц, включающую только набор столбцов с добавлением описаний первичного ключа;
- (b) дополнить описание таблицы реализацией ограничений для описания внешних ключей; для внешних ключей установить свойства контроля целостности данных, если это возможно;
- (c) дополнить описание таблицы реализацией ограничений для описания бизнес-правил;
- (d) дополнить описание таблицы реализацией комментариев для значимых элементов таблицы.

3. Заполнить с помощью SQL-скрипта строками данных для проверки правильного выбора первичных ключей и работоспособности ссылок между таблицами:

- (a) строками данных сначала заполнять мастер-таблицы;
- (b) в каждую таблицу добавить 5 – 10 строк осмысленных данных;
- (c) если не удастся добавить данные в таблицу по причине нарушения уникальности первичного ключа, то следует перепроверить описание этого первичного ключа и его смысл для реального мира;
- (d) если не удастся добавить данные в таблицу по причине нарушения ссылочной целостности, то следует убедиться, что целевые данные существуют, иначе перепроверить описание внешнего ключа.

4. Рассмотреть простые действия по изменению структуры таблицы (переименование столбца таблицы, добавление и удаление ограничений на столбец таблицы или всю таблицу) и реализовать их с помощью оператора ALTER TABLE.

5. Создать временную таблицу с помощью оператора CREATE TABLE и удалить ее с помощью оператора DROP TABLE.

3 ХОД ВЫПОЛНЕНИЯ РАБОТЫ

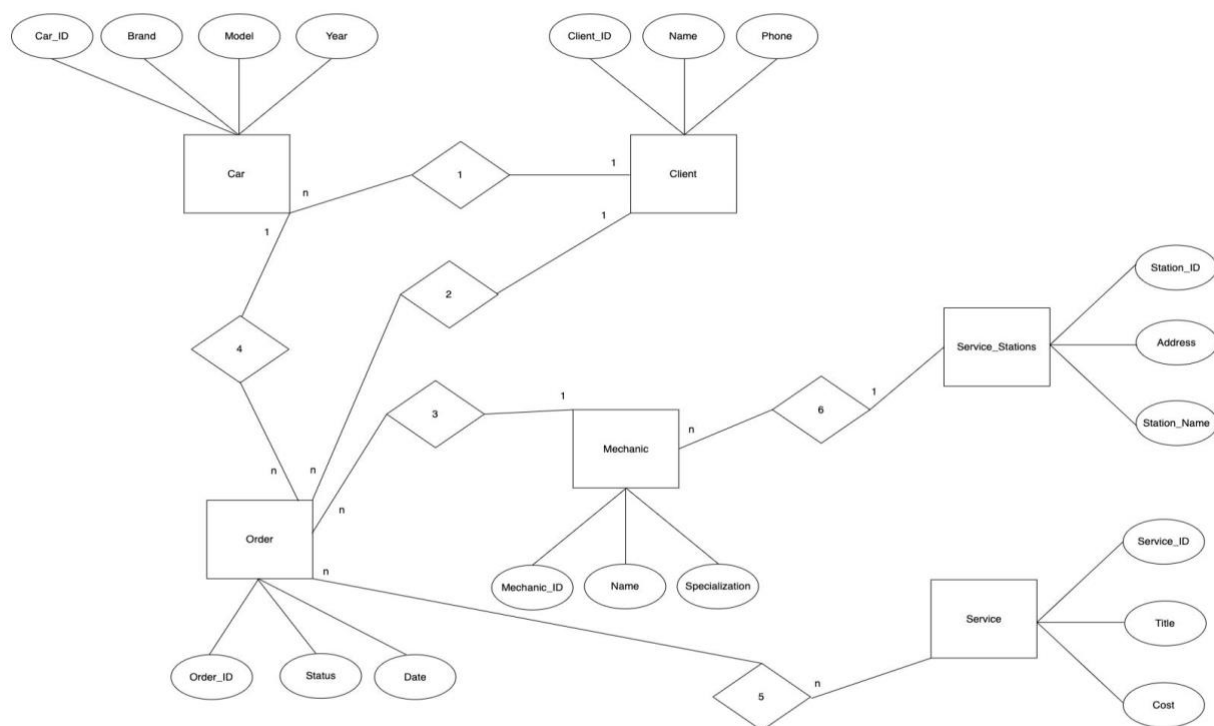


Рисунок 3.1 - ER-диаграмма модели СТО

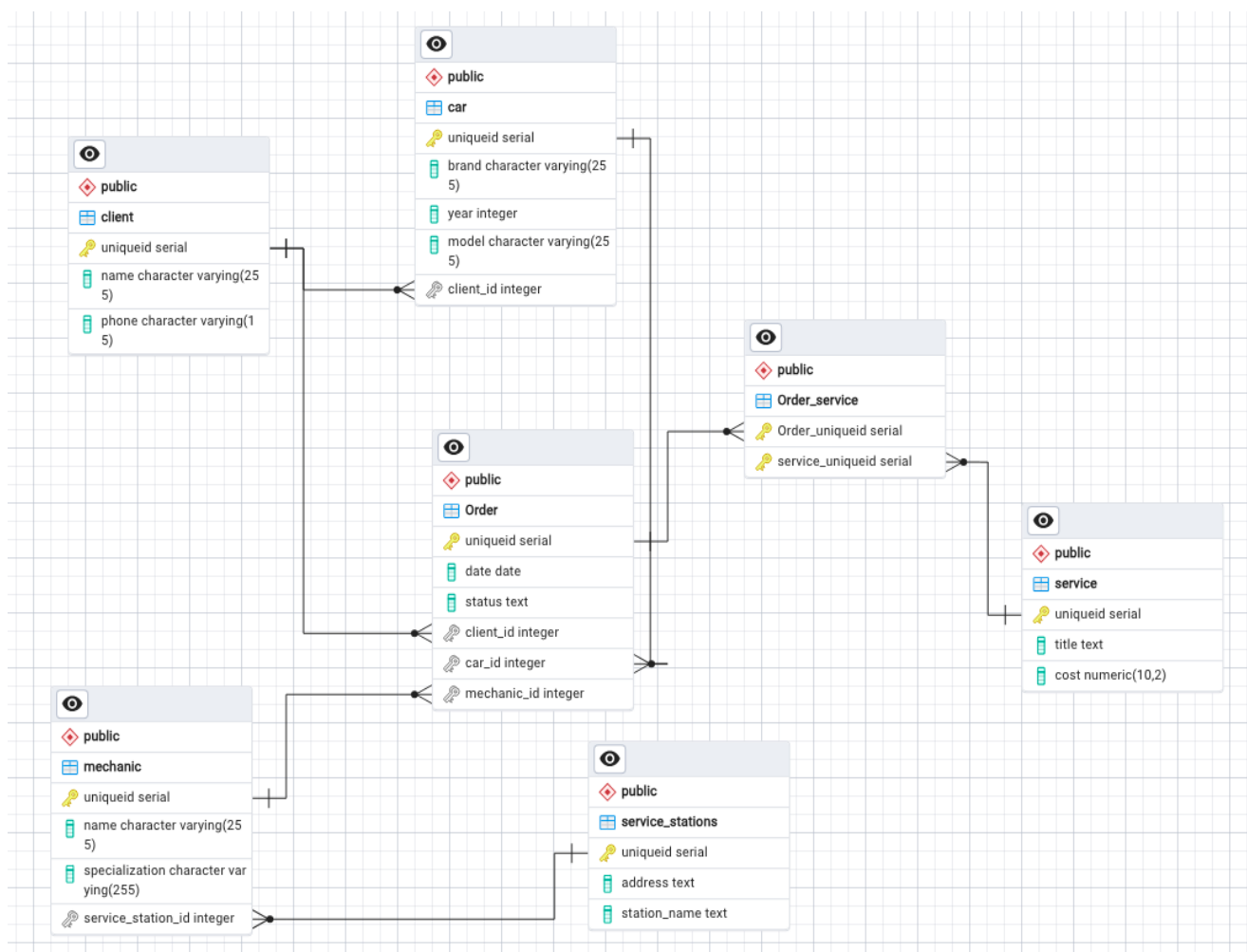


Рисунок 3.2 – Реляционная модель СТО

3.1 Создание таблиц

Ниже представлен листинг скрипта, выполняющего создание таблиц базы данных СТО с помощью команды CREATE TABLE.

```

CREATE TABLE Service_Stations (
    UniqueID SERIAL PRIMARY KEY,
    Address TEXT NOT NULL,
    Station_Name TEXT
);
  
```

```

CREATE TABLE Client (
  
```

```
UniqueID SERIAL PRIMARY KEY,  
Name VARCHAR(255) NOT NULL,  
Phone VARCHAR(15)  
);
```

```
CREATE TABLE Car (  
    UniqueID SERIAL PRIMARY KEY,  
    Brand VARCHAR(255) NOT NULL,  
    Year INT NOT NULL,  
    Model VARCHAR(255) NOT NULL,  
    Client_ID INT,  
    FOREIGN KEY (Client_ID) REFERENCES Client (UniqueID)  
);
```

```
CREATE TABLE Mechanic (  
    UniqueID SERIAL PRIMARY KEY,  
    Name VARCHAR(255) NOT NULL,  
    Specialization VARCHAR(255),  
    Service_Station_ID INT REFERENCES Service_Stations (UniqueID)  
);
```

```
CREATE TABLE Service (  
    UniqueID SERIAL PRIMARY KEY,  
    Title TEXT NOT NULL,  
    Cost DECIMAL (10, 2)  
);
```

```
CREATE TABLE "Order" (  
    UniqueID SERIAL PRIMARY KEY,  
    Date DATE NOT NULL,  
    Status TEXT NOT NULL,
```

```

    Client_ID INT REFERENCES Client (UniqueID),
    Car_ID INT REFERENCES Car (UniqueID),
    Mechanic_ID INT REFERENCES Mechanic (UniqueID)
);

```

```

CREATE TABLE Order_Service (
    Order_ID INT REFERENCES "Order" (UniqueID),
    Service_ID INT REFERENCES Service (UniqueID),
    PRIMARY KEY(Order_ID, Service_ID)
);

```

3.2 Заполнение таблиц

Ниже представлен листинг скрипта, выполняющего заполнение созданных в подразделе 3.2 таблиц базы данных СТО с помощью команды INSERT INTO.

```

INSERT INTO Service (Title, Cost)
VALUES
    ('Tire Rotation', 60.00),
    ('Engine Tune-Up', 100.00),
    ('Wheel Alignment', 70.00),
    ('Transmission Flush', 120.00),
    ('Coolant Flush', 80.00),
    ('Air Filter Replacement', 40.00),
    ('Spark Plug Replacement', 50.00),
    ('Battery Replacement', 90.00),
    ('Alternator Replacement', 150.00),
    ('Fuel Injector Cleaning', 70.00),
    ('Radiator Flush', 60.00),
    ('Timing Belt Replacement', 200.00),
    ('Power Steering Flush', 60.00),

```

```

('Brake Pad Replacement', 100.00),
('Exhaust System Repair', 120.00),
('AC Recharge', 80.00),
('Suspension Repair', 150.00),
('Headlight Restoration', 50.00),
('Windshield Wiper Replacement', 30.00),
('Cabin Air Filter Replacement', 40.00),
('Fuel Filter Replacement', 50.00),
('Oxygen Sensor Replacement', 70.00),
('Starter Replacement', 120.00),
('Thermostat Replacement', 80.00),
('Driveshaft Replacement', 200.00),
('Catalytic Converter Replacement', 250.00),
('Wheel Bearing Replacement', 100.00),
('Steering Rack Replacement', 200.00),
('Clutch Replacement', 300.00);

```

```

INSERT INTO Service_Stations (Address, Station_Name)

```

```

VALUES

```

```

('Pushkin Street, 20', 'AutoCare Center'),
('Moskovskaya Street, 25', 'CarFix Service'),
('Lenina Avenue, 30', 'QuickFix Garage'),
('Revolution Street, 35', 'MasterMechanic Workshop'),
('Industrial Avenue, 40', 'ProAuto Shop'),
('Freedom Square, 45', 'WheelWizard Garage'),
('Victory Boulevard, 50', 'SpeedyService Station'),
('Liberty Lane, 55', 'EliteAuto Care'),
('Sunset Drive, 60', 'SuperiorService Center'),
('Maple Street, 65', 'TopGear Garage'),
('Oak Avenue, 70', 'PrimeAuto Solutions'),
('Greenwood Road, 75', 'AutoMasters Workshop'),

```

```
( 'Birch Lane, 80', 'AceAutomotive Service'),
( 'Riverfront Avenue, 85', 'GoldenWrench Repair'),
( 'Parkview Terrace, 90', 'FirstClass AutoCare'),
( 'Lakeview Drive, 95', 'A1 AutoShop'),
( 'Springfield Avenue, 100', 'ExpertMechanics Garage'),
( 'Main Street, 105', 'AutoTech Center'),
( 'Washington Avenue, 110', 'AutoPro Service'),
( 'Central Park, 115', 'SpeedyFix Garage'),
( 'Cedar Lane, 120', 'QuickFix Solutions'),
( 'Highland Avenue, 125', 'PrecisionAuto Care'),
( 'Rosewood Drive, 130', 'MegaMechanics Workshop'),
( 'Cherry Street, 135', 'StarAuto Repair'),
( 'Sunrise Boulevard, 140', 'TopNotch Garage'),
( 'Pine Street, 145', 'SwiftService Center'),
( 'Forest Drive, 150', 'AutoGenius Workshop'),
( 'Crescent Avenue, 155', 'MasterAuto Care'),
( 'Oakwood Drive, 160', 'GrandAuto Garage');
```

```
INSERT INTO Client (Name, Phone)
```

```
VALUES
```

```
( 'Michael Johnson', '+375291234500'),
( 'Emily Brown', '+375291234501'),
( 'Daniel Davis', '+375291234502'),
( 'Emma Wilson', '+375291234503'),
( 'Matthew Martinez', '+375291234504'),
( 'Olivia Anderson', '+375291234505'),
( 'Christopher Taylor', '+375291234506'),
( 'Ava Thomas', '+375291234507'),
( 'Joshua Moore', '+375291234508'),
( 'Sophia Jackson', '+375291234509'),
( 'William White', '+375291234510'),
```



```
( 'Isabella Harris', '+375291234511'),
( 'James Martin', '+375291234512'),
( 'Charlotte Thompson', '+375291234513'),
( 'John Garcia', '+375291234514'),
( 'Amelia Martinez', '+375291234515'),
( 'Benjamin Lee', '+375291234516'),
( 'Mia Robinson', '+375291234517'),
( 'Ethan Clark', '+375291234518'),
( 'Harper Rodriguez', '+375291234519'),
( 'David Lewis', '+375291234520'),
( 'Madison Walker', '+375291234521'),
( 'Alexander Hall', '+375291234522'),
( 'Elizabeth Allen', '+375291234523'),
( 'Jacob Young', '+375291234524'),
( 'Ella King', '+375291234525'),
( 'Ryan Adams', '+375291234526'),
( 'Avery Wright', '+375291234527'),
( 'Sofia Evans', '+375291234528');
```

```
INSERT INTO "Order" (Date, Status, Client_ID, Car_ID, Mechanic_ID)
VALUES
```

```
( '2024-02-01', 'In Progress', 1, 1, 1),
( '2024-02-02', 'Completed', 2, 2, 2),
( '2024-02-03', 'In Progress', 3, 3, 3),
( '2024-02-04', 'Completed', 4, 4, 4),
( '2024-02-05', 'In Progress', 5, 5, 5),
( '2024-02-06', 'Completed', 6, 6, 6),
( '2024-02-07', 'In Progress', 7, 7, 7),
( '2024-02-08', 'Completed', 8, 8, 8),
( '2024-02-09', 'In Progress', 9, 9, 9),
( '2024-02-10', 'Completed', 10, 10, 10),
```

```

('2024-02-11', 'In Progress', 11, 11, 11),
('2024-02-12', 'Completed', 12, 12, 12),
('2024-02-13', 'In Progress', 13, 13, 13),
('2024-02-14', 'Completed', 14, 14, 14),
('2024-02-15', 'In Progress', 15, 15, 15),
('2024-02-16', 'Completed', 16, 16, 16),
('2024-02-17', 'In Progress', 17, 17, 17),
('2024-02-18', 'Completed', 18, 18, 18),
('2024-02-19', 'In Progress', 19, 19, 19),
('2024-02-20', 'Completed', 20, 20, 20),
('2024-02-21', 'In Progress', 21, 21, 21),
('2024-02-22', 'Completed', 22, 22, 22),
('2024-02-23', 'In Progress', 23, 23, 23),
('2024-02-24', 'Completed', 24, 24, 24),
('2024-02-25', 'In Progress', 25, 25, 25),
('2024-02-26', 'Completed', 26, 26, 26),
('2024-02-27', 'In Progress', 27, 27, 27),
('2024-02-28', 'Completed', 28, 28, 28),
('2024-02-29', 'In Progress', 29, 29, 29),
('2024-03-01', 'Completed', 30, 30, 30);

```

```

INSERT INTO Car (Brand, Year, Model, Client_ID)
VALUES

```

```

('Toyota', 2018, 'Camry', 1),
('Honda', 2019, 'Civic', 2),
('Ford', 2017, 'Focus', 3),
('Chevrolet', 2016, 'Malibu', 4),
('Nissan', 2020, 'Altima', 5),
('Subaru', 2015, 'Outback', 6),
('Volkswagen', 2018, 'Jetta', 7),
('Hyundai', 2019, 'Elantra', 8),

```

```
( 'BMW', 2017, '3 Series', 9),
( 'Mercedes-Benz', 2016, 'C-Class', 10),
( 'Audi', 2020, 'A4', 11),
( 'Lexus', 2015, 'ES', 12),
( 'Kia', 2018, 'Optima', 13),
( 'Mazda', 2019, 'Mazda3', 14),
( 'Jeep', 2017, 'Cherokee', 15),
( 'Ram', 2016, '1500', 16),
( 'GMC', 2020, 'Sierra', 17),
( 'Buick', 2015, 'Enclave', 18),
( 'Chrysler', 2018, '300', 19),
( 'Lincoln', 2019, 'MKZ', 20),
( 'Cadillac', 2017, 'XT5', 21),
( 'Volvo', 2016, 'XC60', 22),
( 'Tesla', 2020, 'Model S', 23),
( 'Fiat', 2015, '500', 24),
( 'Jaguar', 2018, 'XE', 25),
( 'Land Rover', 2019, 'Discovery Sport', 26),
( 'Porsche', 2017, 'Macan', 27),
( 'Infiniti', 2016, 'Q50', 28),
( 'Mini', 2020, 'Cooper', 29),
( 'Acura', 2015, 'TLX', 30);
```

```
INSERT INTO Mechanic (Name, Specialization, Service_Station_ID)
VALUES
```

```
( 'John Smith', 'Engine Specialist', 60),
( 'Emma Johnson', 'Transmission Expert', 61),
( 'Michael Williams', 'Brake Technician', 62),
( 'Emily Brown', 'Electrical Systems', 63),
( 'Christopher Davis', 'Suspension Specialist', 64),
( 'Sarah Wilson', 'Air Conditioning', 65),
```

```

('David Jones', 'Fuel System Technician', 66),
('Jessica Martinez', 'Diagnostician', 67),
('Ryan Taylor', 'Alignment Specialist', 68),
('Ashley Garcia', 'Exhaust Systems', 69),
('Matthew Rodriguez', 'Oil Change Expert', 70),
('Jennifer Lopez', 'Cooling System Technician', 71),
('Daniel Lee', 'Steering System Technician', 72),
('Lisa Moore', 'Tire Technician', 73),
('Robert Jackson', 'Engine Tuner', 74),
('Elizabeth White', 'Electronics Specialist', 75),
('William Harris', 'Brake Repair Technician', 76),
('Samantha Hall', 'Transmission Rebuilder', 77),
('Christopher Walker', 'HVAC Specialist', 78),
('Laura Young', 'Suspension Tuner', 79),
('Kevin Clark', 'Fuel Injection Specialist', 80),
('Amanda Hill', 'Diagnostic Technician', 81),
('James Baker', 'Alignment Technician', 82),
('Sarah Carter', 'Exhaust System Specialist', 83),
('Nicholas Thomas', 'Oil Change Technician', 84),
('Rebecca Allen', 'Cooling System Expert', 85),
('Justin Lewis', 'Steering System Specialist', 86),
('Stephanie King', 'Tire Technician', 87),
('Andrew Scott', 'Engine Performance Tuner', 88),
('Megan Green', 'Electrical Systems Specialist', 89);

```

```

INSERT INTO "Order" (Date, Status, Client_ID, Car_ID, Mechanic_ID)
VALUES

```

```

('2024-02-01', 'In Progress', 1, 31, 31),
('2024-02-02', 'Completed', 2, 32, 32),
('2024-02-03', 'In Progress', 3, 33, 33),
('2024-02-04', 'Completed', 4, 34, 34),

```

```

('2024-02-05', 'In Progress', 5, 35, 35),
('2024-02-06', 'Completed', 6, 36, 36),
('2024-02-07', 'In Progress', 7, 37, 37),
('2024-02-08', 'Completed', 8, 38, 38),
('2024-02-09', 'In Progress', 9, 39, 39),
('2024-02-10', 'Completed', 10, 40, 40),
('2024-02-11', 'In Progress', 11, 41, 41),
('2024-02-12', 'Completed', 12, 42, 42),
('2024-02-13', 'In Progress', 13, 43, 43),
('2024-02-14', 'Completed', 14, 44, 44),
('2024-02-15', 'In Progress', 15, 45, 45),
('2024-02-16', 'Completed', 16, 46, 46),
('2024-02-17', 'In Progress', 17, 47, 47),
('2024-02-18', 'Completed', 18, 48, 48),
('2024-02-19', 'In Progress', 19, 49, 49),
('2024-02-20', 'Completed', 20, 50, 50),
('2024-02-21', 'In Progress', 21, 51, 51),
('2024-02-22', 'Completed', 22, 52, 52),
('2024-02-23', 'In Progress', 23, 53, 53),
('2024-02-24', 'Completed', 24, 54, 54),
('2024-02-25', 'In Progress', 25, 55, 55),
('2024-02-26', 'Completed', 26, 56, 56),
('2024-02-27', 'In Progress', 27, 57, 57),
('2024-02-28', 'Completed', 28, 58, 58),
('2024-02-29', 'In Progress', 29, 59, 59),
('2024-03-01', 'Completed', 30, 60, 60);

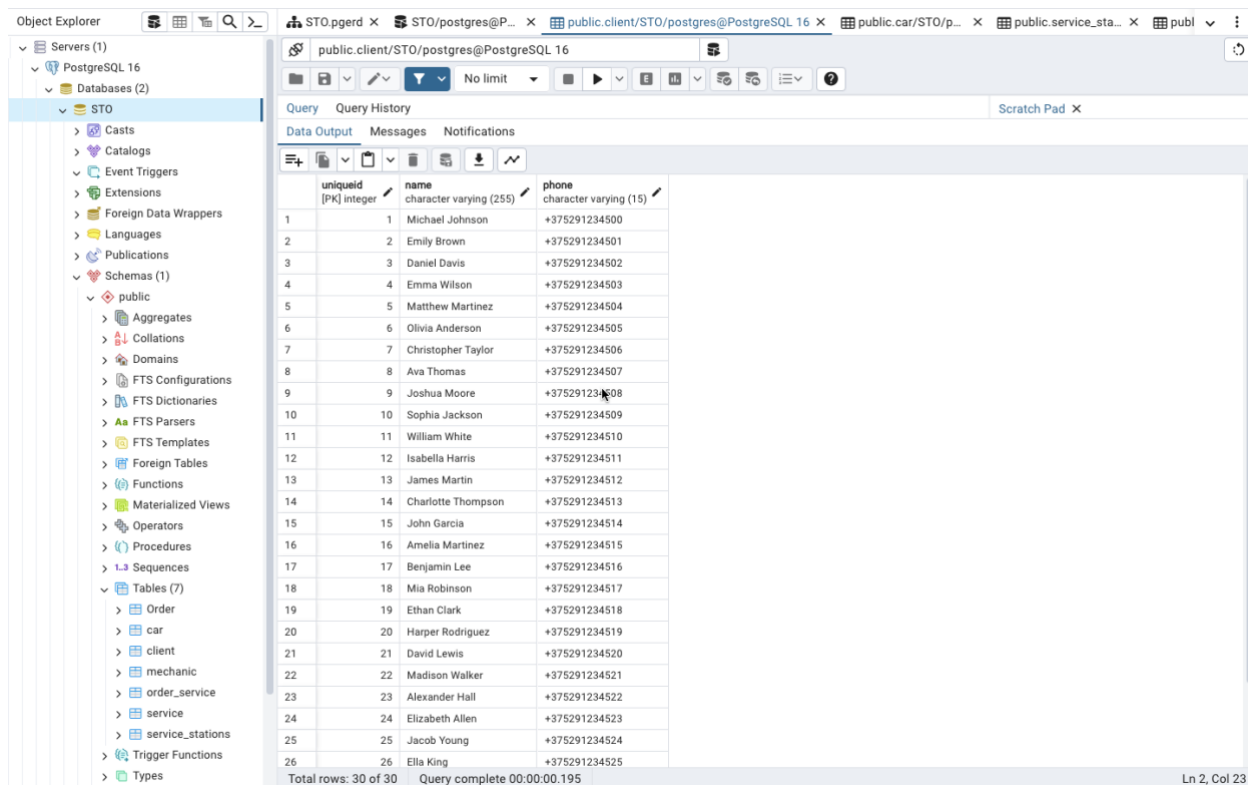
```

```

INSERT INTO Order_Service (Order_ID, Service_ID)
SELECT "Order".UniqueID, Service.UniqueID
FROM "Order"
CROSS JOIN Service;

```

3.3 Демонстрация заполненных таблиц



The screenshot shows a PostgreSQL client interface with a table containing 30 rows of data. The table has three columns: 'uniqueid' (integer, primary key), 'name' (character varying (255)), and 'phone' (character varying (15)). The data is displayed in a grid format with row numbers 1 through 30. The status bar at the bottom indicates 'Total rows: 30 of 30' and 'Query complete 00:00:00.195'.

uniqueid [PK] integer	name character varying (255)	phone character varying (15)
1	Michael Johnson	+375291234500
2	Emily Brown	+375291234501
3	Daniel Davis	+375291234502
4	Emma Wilson	+375291234503
5	Matthew Martinez	+375291234504
6	Olivia Anderson	+375291234505
7	Christopher Taylor	+375291234506
8	Ava Thomas	+375291234507
9	Joshua Moore	+375291234508
10	Sophia Jackson	+375291234509
11	William White	+375291234510
12	Isabella Harris	+375291234511
13	James Martin	+375291234512
14	Charlotte Thompson	+375291234513
15	John Garcia	+375291234514
16	Amelia Martinez	+375291234515
17	Benjamin Lee	+375291234516
18	Mia Robinson	+375291234517
19	Ethan Clark	+375291234518
20	Harper Rodriguez	+375291234519
21	David Lewis	+375291234520
22	Madison Walker	+375291234521
23	Alexander Hall	+375291234522
24	Elizabeth Allen	+375291234523
25	Jacob Young	+375291234524
26	Ella King	+375291234525

Рисунок 3.3 – таблица Client

Object Explorer

public.car/STO/postgres@PostgreSQL 16

Data Output

	uniqueid [PK] integer	brand character varying (255)	year integer	model character varying (255)	client_id integer
1	31	Toyota	2018	Camry	1
2	32	Honda	2019	Civic	2
3	33	Ford	2017	Focus	3
4	34	Chevrolet	2016	Malibu	4
5	35	Nissan	2020	Altima	5
6	36	Subaru	2015	Outback	6
7	37	Volkswagen	2018	Jetta	7
8	38	Hyundai	2019	Elantra	8
9	39	BMW	2017	3 Series	9
10	40	Mercedes-Benz	2016	C-Class	10
11	41	Audi	2020	A4	11
12	42	Lexus	2015	ES	12
13	43	Kia	2018	Optima	13
14	44	Mazda	2019	Mazda3	14
15	45	Jeep	2017	Cherokee	15
16	46	Ram	2016	1500	16
17	47	GMC	2020	Sierra	17
18	48	Buick	2015	Enclave	18
19	49	Chrysler	2018	300	19
20	50	Lincoln	2019	MKZ	20
21	51	Cadillac	2017	XT5	21
22	52	Volvo	2016	XC60	22
23	53	Tesla	2020	Model S	23
24	54	Fiat	2015	500	24
25	55	Jaguar	2018	XE	25
26	56	Land Rover	2019	Discovery Sport	26
27	57	Porsche	2017	Macan	27

Total rows: 30 of 30 Query complete 00:00:00.223 Ln 2, Col 23

Рисунок 3.4 – таблица Car

Object Explorer

public.service_stations/STO/postgres@PostgreSQL 16

Data Output

	uniqueid [PK] integer	address text	station_name text
1	60	Pushkin Street, 20	AutoCare Center
2	61	Moskovskaya Street, 25	CarFix Service
3	62	Lenina Avenue, 30	QuickFix Garage
4	63	Revolution Street, 35	MasterMechanic Workshop
5	64	Industrial Avenue, 40	ProAuto Shop
6	65	Freedom Square, 45	WheelWizard Garage
7	66	Victory Boulevard, 50	SpeedyService Station
8	67	Liberty Lane, 55	EliteAuto Care
9	68	Sunset Drive, 60	SuperiorService Center
10	69	Maple Street, 65	TopGear Garage
11	70	Oak Avenue, 70	PrimeAuto Solutions
12	71	Greenwood Road, 75	AutoMasters Workshop
13	72	Birch Lane, 80	AceAutomotive Service
14	73	Riverfront Avenue, 85	GoldenWrench Repair
15	74	Parkview Terrace, 90	FirstClass AutoCare
16	75	Lakeview Drive, 95	A1 AutoShop
17	76	Springfield Avenue, 100	ExpertMechanics Garage
18	77	Main Street, 105	AutoTech Center
19	78	Washington Avenue, 1...	AutoPro Service
20	79	Central Park, 115	SpeedyFix Garage
21	80	Cedar Lane, 120	QuickFix Solutions
22	81	Highland Avenue, 125	PrecisionAuto Care
23	82	Rosewood Drive, 130	MegaMechanics Workshop
24	83	Cherry Street, 135	StarAuto Repair
25	84	Sunrise Boulevard, 140	TopNotch Garage
26	85	Pine Street, 145	SwiftService Center
27	86	Forest Drive, 150	AutoGenius Workshop

Total rows: 30 of 30 Query complete 00:00:00.249 Ln 1, Col 1

Рисунок 3.5 – таблица Service_Stations

Object Explorer

ic.client/STO/... x public.car/STO/p... x public.service_sta... x public.service/STO/postgres@PostgreSQL 16 x public.mechanic/... x put v

public.service/STO/postgres@PostgreSQL 16

Data Output Messages Notifications

uniqueid [PK] integer	title text	cost numeric (10,2)
1	1 Tire Rotation	60.00
2	2 Engine Tune-Up	100.00
3	3 Wheel Alignment	70.00
4	4 Transmission Flush	120.00
5	5 Coolant Flush	80.00
6	6 Air Filter Replacement	40.00
7	7 Spark Plug Replacement	50.00
8	8 Battery Replacement	90.00
9	9 Alternator Replacement	150.00
10	10 Fuel Injector Cleaning	70.00
11	11 Radiator Flush	60.00
12	12 Timing Belt Replacement	200.00
13	13 Power Steering Flush	60.00
14	14 Brake Pad Replacement	100.00
15	15 Exhaust System Repair	120.00
16	16 AC Recharge	80.00
17	17 Suspension Repair	150.00
18	18 Headlight Restoration	50.00
19	19 Windshield Wiper Replacement	30.00
20	20 Cabin Air Filter Replacement	40.00
21	21 Fuel Filter Replacement	50.00
22	22 Oxygen Sensor Replacement	70.00
23	23 Starter Replacement	120.00
24	24 Thermostat Replacement	80.00
25	25 Driveshaft Replacement	200.00
26	26 Catalytic Converter Replacement	250.00
27	27 Wheel Bearing Replacement	100.00

Total rows: 30 of 30 Query complete 00:00:00.681 Ln 1, Col 1

Рисунок 3.6 – таблица Service

Object Explorer

ic.client/STO/... x public.car/STO/p... x public.service_sta... x public.service/ST... x public.mechanic/STO/postgres@PostgreSQL 16 x

public.mechanic/STO/postgres@PostgreSQL 16

Data Output Messages Notifications

uniqueid [PK] integer	name character varying (255)	specialization character varying (255)	service_station_id integer
1	31 John Smith	Engine Specialist	60
2	32 Emma Johnson	Transmission Expert	61
3	33 Michael Williams	Brake Technician	62
4	34 Emily Brown	Electrical Systems	63
5	35 Christopher Davis	Suspension Specialist	64
6	36 Sarah Wilson	Air Conditioning	65
7	37 David Jones	Fuel System Technician	66
8	38 Jessica Martinez	Diagnostician	67
9	39 Ryan Taylor	Alignment Specialist	68
10	40 Ashley Garcia	Exhaust Systems	69
11	41 Matthew Rodriguez	Oil Change Expert	70
12	42 Jennifer Lopez	Cooling System Technician	71
13	43 Daniel Lee	Steering System Technician	72
14	44 Lisa Moore	Tire Technician	73
15	45 Robert Jackson	Engine Tuner	74
16	46 Elizabeth White	Electronics Specialist	75
17	47 William Harris	Brake Repair Technician	76
18	48 Samantha Hall	Transmission Rebuilder	77
19	49 Christopher Walker	HVAC Specialist	78
20	50 Laura Young	Suspension Tuner	79
21	51 Kevin Clark	Fuel Injection Specialist	80
22	52 Amanda Hill	Diagnostic Technician	81
23	53 James Baker	Alignment Technician	82
24	54 Sarah Carter	Exhaust System Specialist	83
25	55 Nicholas Thomas	Oil Change Technician	84
26	56 Rebecca Allen	Cooling System Expert	85
27	57 Justin Lewis	Steering System Specialist	86

Total rows: 30 of 30 Query complete 00:00:00.265 Ln 1, Col 1

Рисунок 3.7 – таблица Mechanic

Object Explorer

public.service_sta... public.service/ST... public.mechanic/... public.Order/STO/postgres@PostgreSQL 16 public.order_servi...

public.Order/STO/postgres@PostgreSQL 16

Data Output Messages Notifications

	uniqueid [PK] integer	date date	status text	client_id integer	car_id integer	mechanic_id integer
1	1	2024-02-01	In Progress	1	31	31
2	2	2024-02-02	Completed	2	32	32
3	3	2024-02-03	In Progress	3	33	33
4	4	2024-02-04	Completed	4	34	34
5	5	2024-02-05	In Progress	5	35	35
6	6	2024-02-06	Completed	6	36	36
7	7	2024-02-07	In Progress	7	37	37
8	8	2024-02-08	Completed	8	38	38
9	9	2024-02-09	In Progress	9	39	39
10	10	2024-02-10	Completed	10	40	40
11	11	2024-02-11	In Progress	11	41	41
12	12	2024-02-12	Completed	12	42	42
13	13	2024-02-13	In Progress	13	43	43
14	14	2024-02-14	Completed	14	44	44
15	15	2024-02-15	In Progress	15	45	45
16	16	2024-02-16	Completed	16	46	46
17	17	2024-02-17	In Progress	17	47	47
18	18	2024-02-18	Completed	18	48	48
19	19	2024-02-19	In Progress	19	49	49
20	20	2024-02-20	Completed	20	50	50
21	21	2024-02-21	In Progress	21	51	51
22	22	2024-02-22	Completed	22	52	52
23	23	2024-02-23	In Progress	23	53	53
24	24	2024-02-24	Completed	24	54	54
25	25	2024-02-25	In Progress	25	55	55
26	26	2024-02-26	Completed	26	56	56
27	27	2024-02-27	In Progress	27	57	57

Total rows: 30 of 30 Query complete 00:00:00.308 Ln 1, Col 1

Рисунок 3.8 – таблица Order

Object Explorer

public.service_sta... public.service/ST... public.mechanic/... public.Order/STO/... public.order_service/STO/postgres@PostgreSQL 16

public.order_service/STO/postgres@PostgreSQL 16

Data Output Messages Notifications

	order_id [PK] integer	service_id [PK] integer
1	1	1
2	1	2
3	1	3
4	1	4
5	1	5
6	1	6
7	1	7
8	1	8
9	1	9
10	1	10
11	1	11
12	1	12
13	1	13
14	1	14
15	1	15
16	1	16
17	1	17
18	1	18
19	1	19
20	1	20
21	1	21
22	1	22
23	1	23
24	1	24
25	1	25
26	1	26
27	1	27

Total rows: 900 of 900 Query complete 00:00:00.304 Ln 2, Col 39

Рисунок 3.9 – таблица Order_Service

